RESEARCH PORTFOLIO

Robotic Mechanisms, Actuation & System Integration

Shuwei Zhao

B.Eng. Candidate in Electronic Engineering
The Chinese University of Hong Kong
Expected graduation: May 2027

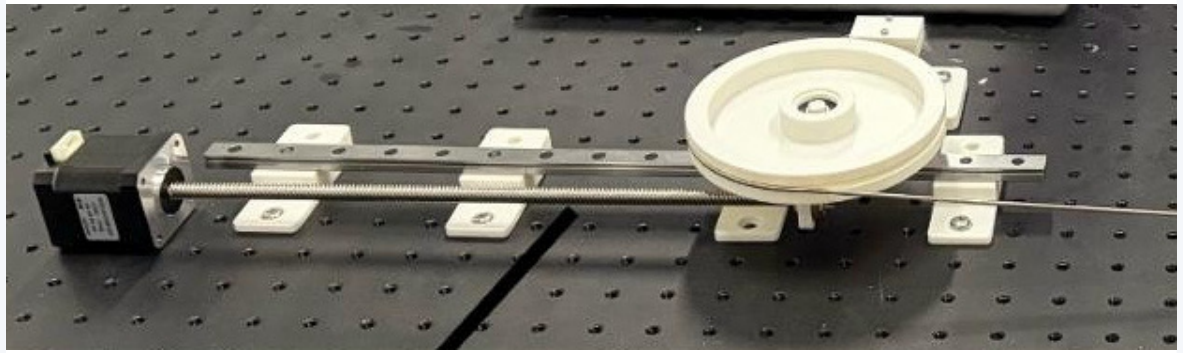
Selected projects in mechanical design, embedded control, magnetic and soft actuation, and experimental validation.

RESEARCH INTERESTS

Cable-driven mechanisms, compliant robotics, and wearable systems.

PHD APPLICATIONS

Fall 2027
Hardware development - actuation - sensing - system integration



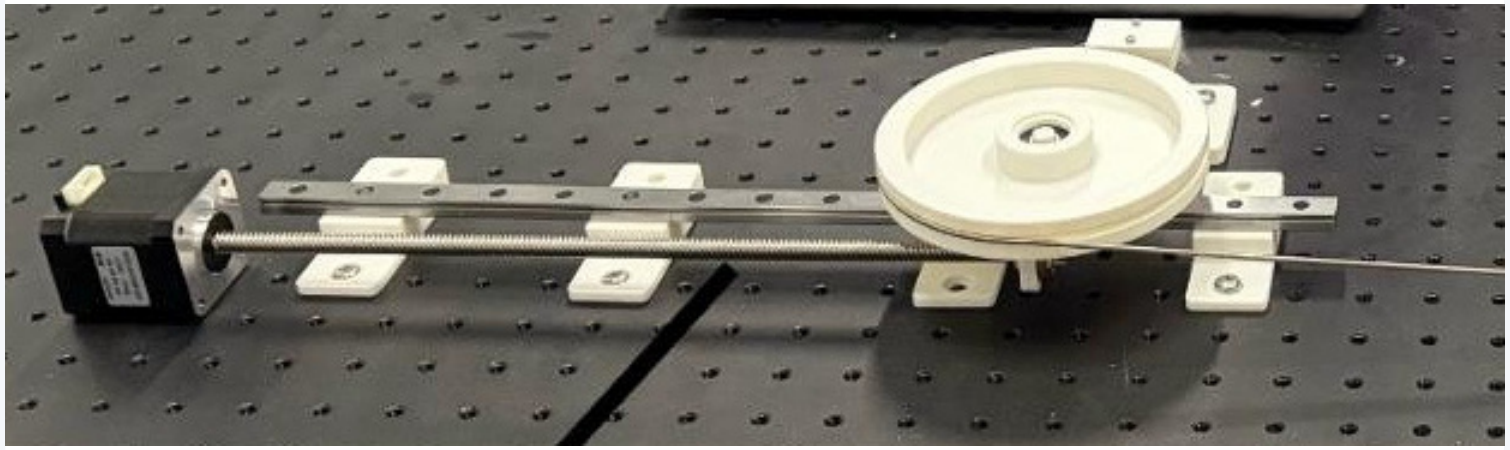
My contribution at XJTU: the translating-pulley catheter accumulator, mechanical assembly, Arduino firmware, and PC motion-control interface. Real photograph, cropped.

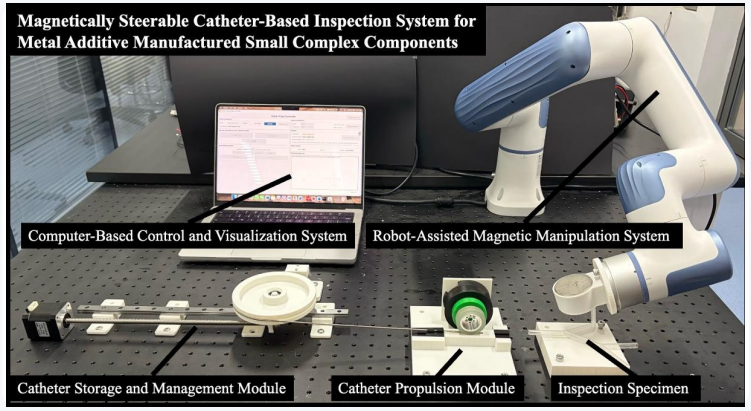
CUHK | Xi'an Jiaotong University | ETH Zürich

PROJECT 01 | XJTU | JULY-AUGUST 2026

Motorized Catheter Management for Industrial Inspection

Research Assistant | Prof. Qiji Ze's group, Xi'an Jiaotong University





Laboratory platform: robot-assisted magnetic manipulation and the complete industrial-inspection system were team infrastructure. The original photograph provides system context.

MECHANISM DESIGN

Designed and built a translating-pulley catheter accumulator; approximately $\Delta L \approx 2\Delta x$ length compensation. Fusion 360 / SolidWorks assembly with a 120-mm guide pulley, 608 bearings, MGN12H carriage, T8 lead screw, and 3D-printed interfaces.

COMPLETE CONTROL CHAIN

Python/Tkinter → USB serial → Arduino UNO → TB6600 → stepper-driven stage. Calibration, jogging, fixed-distance motion, adjustable speed/acceleration, travel limits, and a communication watchdog.

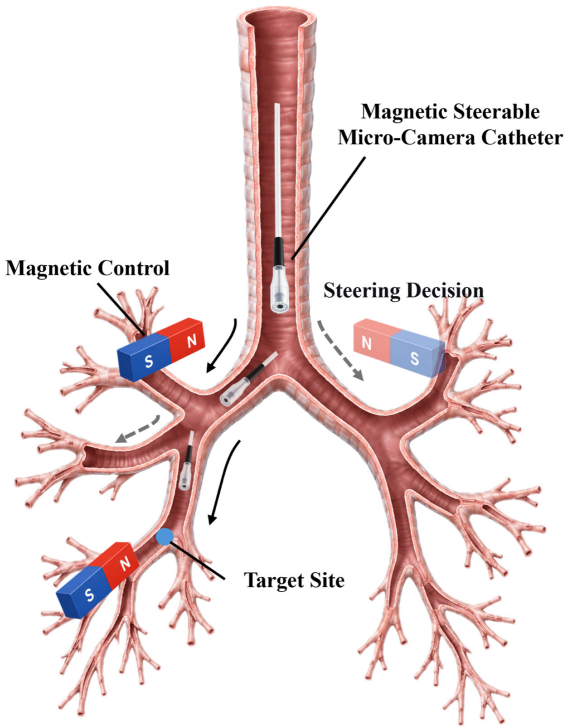
Foundation for further work in robotic mechanisms, motion transmission, and actuation systems.

My contribution: the translating-pulley catheter accumulator, its mechanical assembly, Arduino firmware, and PC motion-control interface.

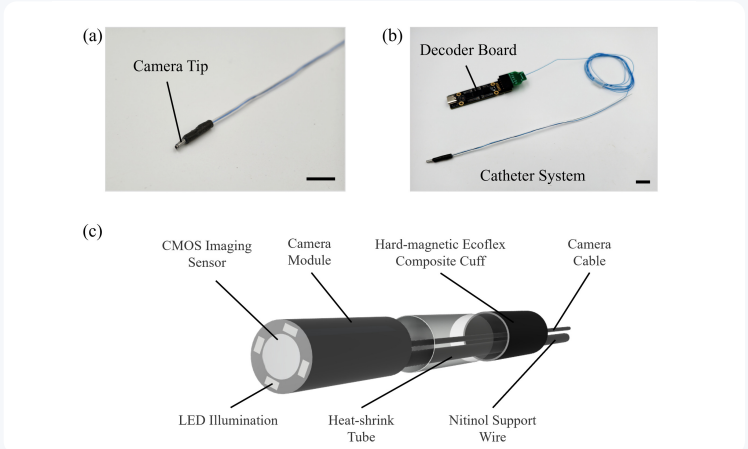
PROJECT 02 | CUHK | JUNE-JULY 2026

Magnetically Steerable Micro-Camera Catheter

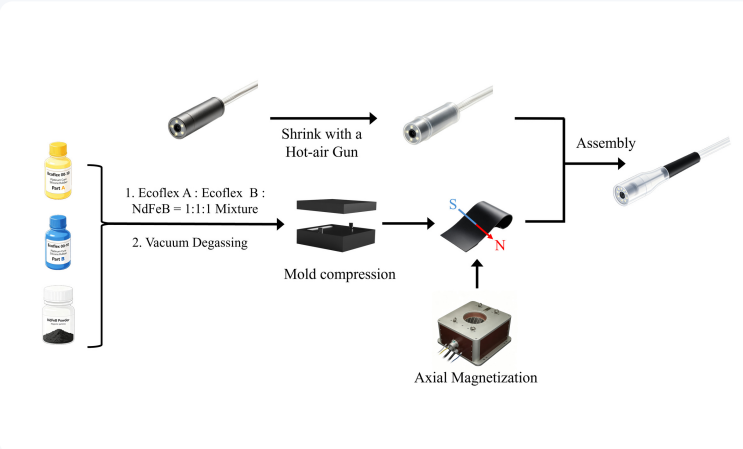
Co-first author | Accepted at ICBIR 2026 | Advisor: Prof. Hongliang Ren



Design concept for bronchial-tree phantom navigation



Prototype structure



Composite preparation, magnetization, and distal assembly

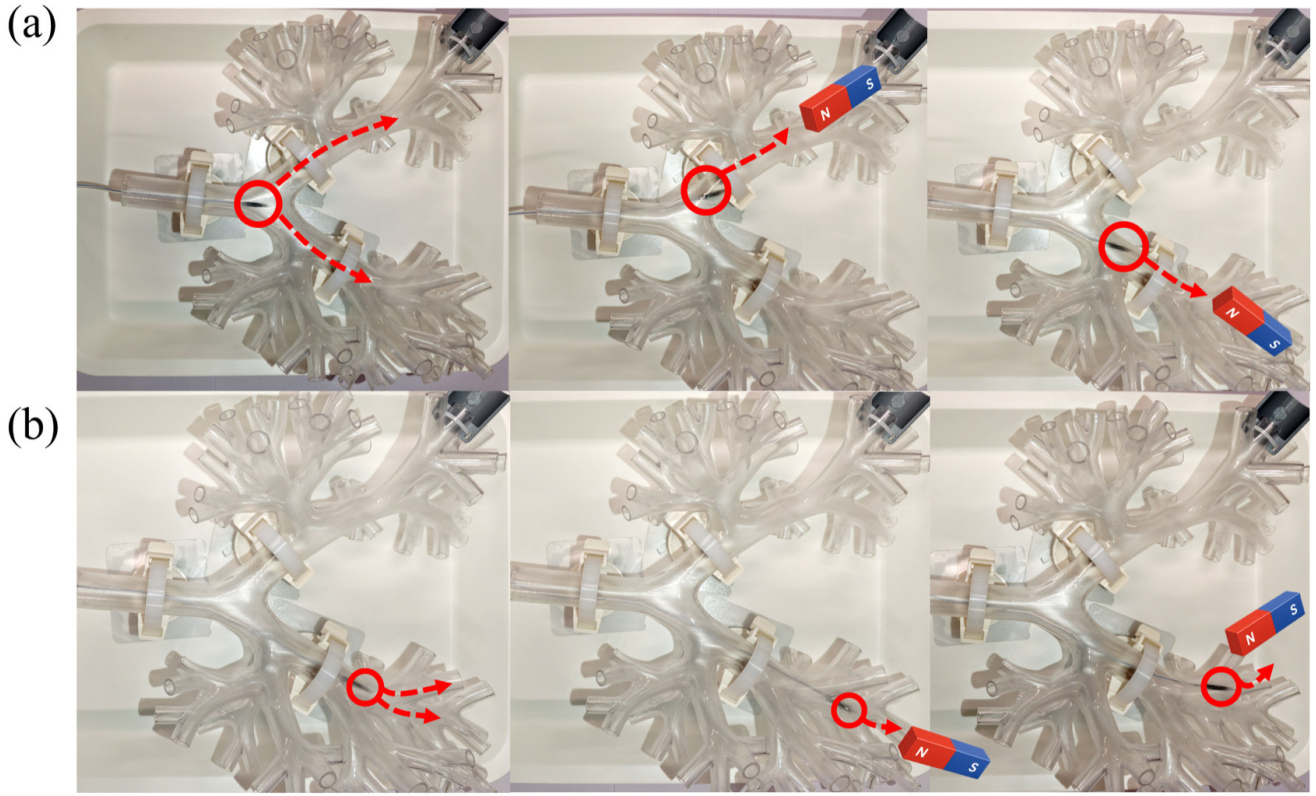
MY CONTRIBUTION

Designed and fabricated the millimeter-scale catheter; integrated the LED micro-camera, 0.2-mm nitinol support wire, heat-shrink encapsulation, and hard-magnetic Ecoflex cuff. Developed insertion-force and steering-characterization setups and contributed to manuscript and figure preparation.

Shuwei Zhao†, Ruizhou Zhao†, Zhiqing Tang, Wenchao Yue, Hongliang Ren. †Shuwei Zhao and Ruizhou Zhao contributed equally.

Magnetic Steering and Forward-Looking Imaging

Bronchial-tree phantom experiments | Manual external permanent-magnet actuation



Magnetically assisted branch redirection. Image sequence reproduced from existing experimental materials.

INSERTION CHARACTERIZATION

Characterized insertion force under dry and saline-wetted conditions.

STEERING AND NAVIGATION

Evaluated multidirectional magnetic steering and forward-looking imaging in a bronchial-tree phantom, including primary and secondary bifurcations.

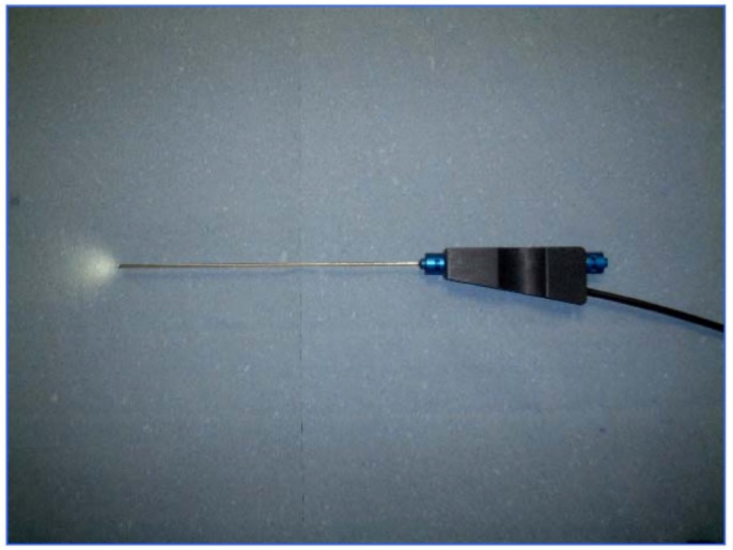
SCOPE OF EVIDENCE

Benchtop phantom evaluation. Further work includes controlled magnetic actuation, visual feedback, and repeated quantitative navigation tests.

PROJECT 03 | ETH ZÜRICH | MARCH-JUNE 2026

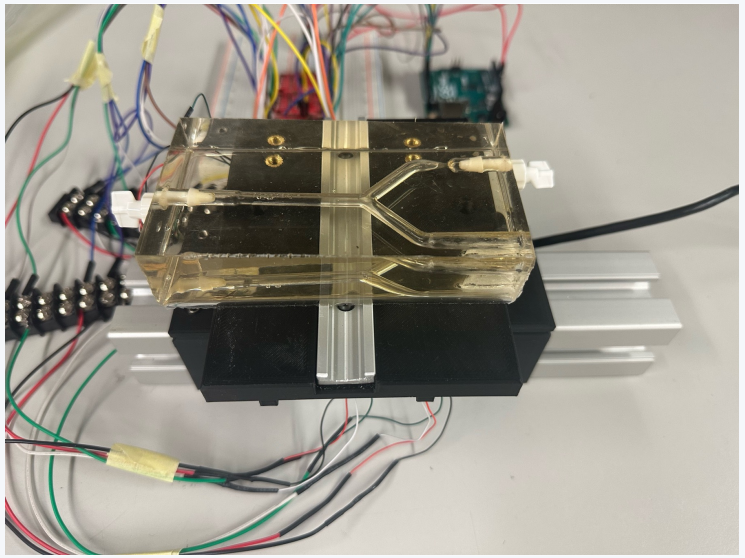
Soft Miniaturized Intraductal Imaging Catheter

Multi-Scale Robotics Lab | Prof. Bradley Nelson | Semester-project grade: 5.5/6.0



FLEXIBLE PROTOTYPES

Designed and fabricated flexible imaging catheter prototypes using 0.5-mm and 0.7-mm nitinol support wires, investigating flexibility, pushability, and insertion resistance. These are support-wire diameters.



CALIBRATED FORCE MEASUREMENT

Built and calibrated a four-load-cell platform with Arduino/Python acquisition. The reported analysis used 10 selected trials from 20 recorded manual trials per condition and included a 1.1-mm rigid reference.

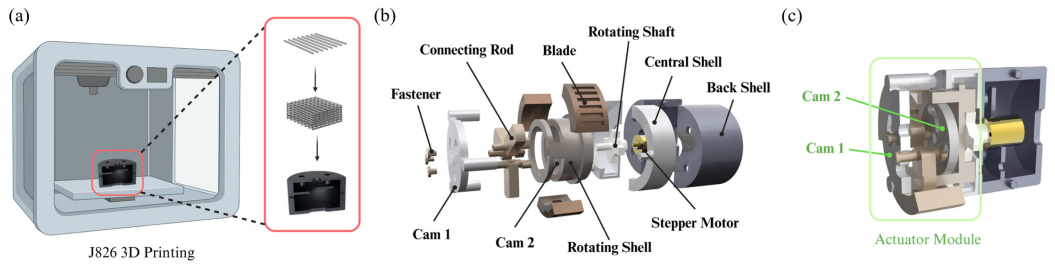


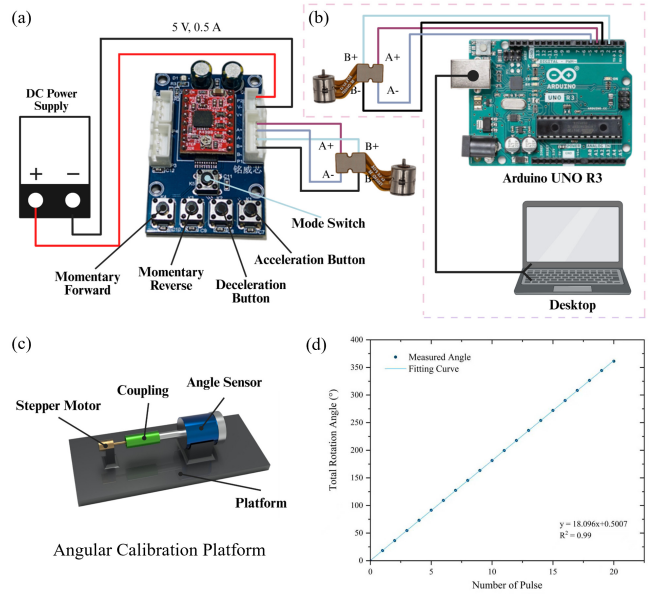
EXPERIMENTAL COMPARISON

Compared support and surface configurations. Fabricated a silicone breast-duct phantom and evaluated imaging with external illumination, including a porcine-blood condition. Results are limited to simplified benchtop models.

Kirigami Capsule for Gastrointestinal Biopsy

Contributing author | Accepted at IEEE ICRA 2026 | Advisor: Prof. Hongliang Ren





SYSTEM INTEGRATION AND EX VIVO VALIDATION

Contributed to iterative assembly, ex vivo porcine experiments, quantitative analysis, and validation writing. Team prototype: 17-mm diameter and 22-mm length.

MOTOR CONTROL AND ANGULAR CALIBRATION

Developed motor-control electronics and Arduino pulse-control logic. Built an angular calibration platform (approximately 18.1 degrees per pulse; $R^2 = 0.99$).

Ruizhou Zhao†, Yichen Chu†, Shuwei Zhao, Wenchao Yue, Raymond Shing-Yan Tang, Hongliang Ren. †Ruizhou Zhao and Yichen Chu contributed equally.

Author manuscript: arxiv.org/abs/2602.06207. Ex vivo prototype evaluation does not establish clinical safety.

From Hardware Integration to Robotic Assistance

PNEUMATIC SOFT JUMPING ROBOT

Designed and fabricated a double-layer spherical-cap silicone actuator and Arduino pneumatic control for benchtop jumping. Developed geometric models and investigated actuator radius and opening angle, deformation, and elastic-energy storage. Ongoing since September 2025.

EXISTING WORK | IMAGING AND DELIVERY CATHETER

My current work focuses on integrating the imaging module, magnetic steering components, and an independent drug-delivery channel within a flexible catheter. Prototype development is ongoing, and experimental validation is planned.

INTENDED PHD RESEARCH DIRECTIONS

I am interested in extending my experience in mechanism design, actuation, and embedded control toward cable-driven and wearable robotic systems. Potential directions include modular actuation interfaces, lightweight compliant mechanisms, and sensing and control for physical assistance.

These are intended PhD research directions rather than completed projects.

I aim to deepen my expertise in modeling, control, and learning-based methods while building on my hardware and system-integration experience.